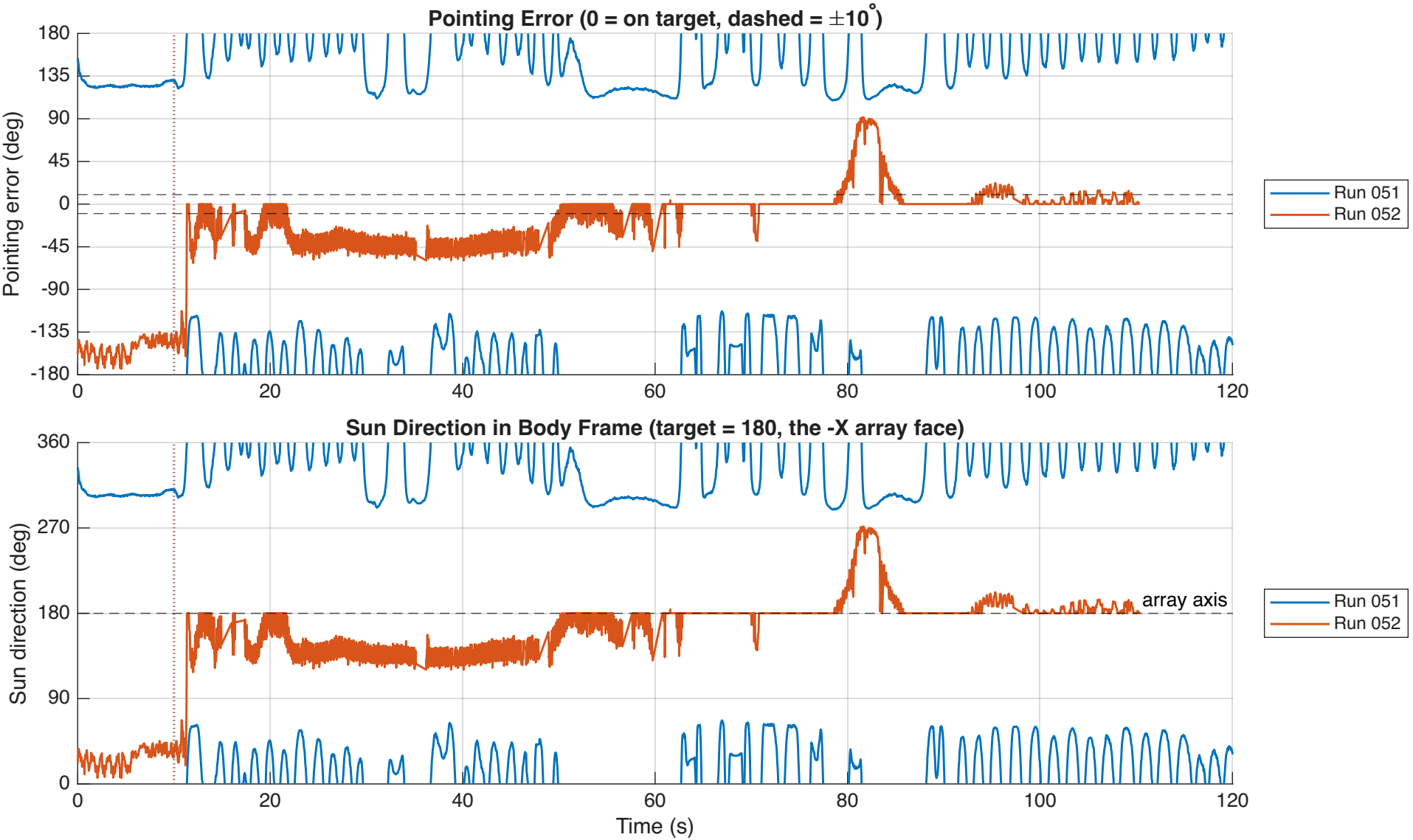
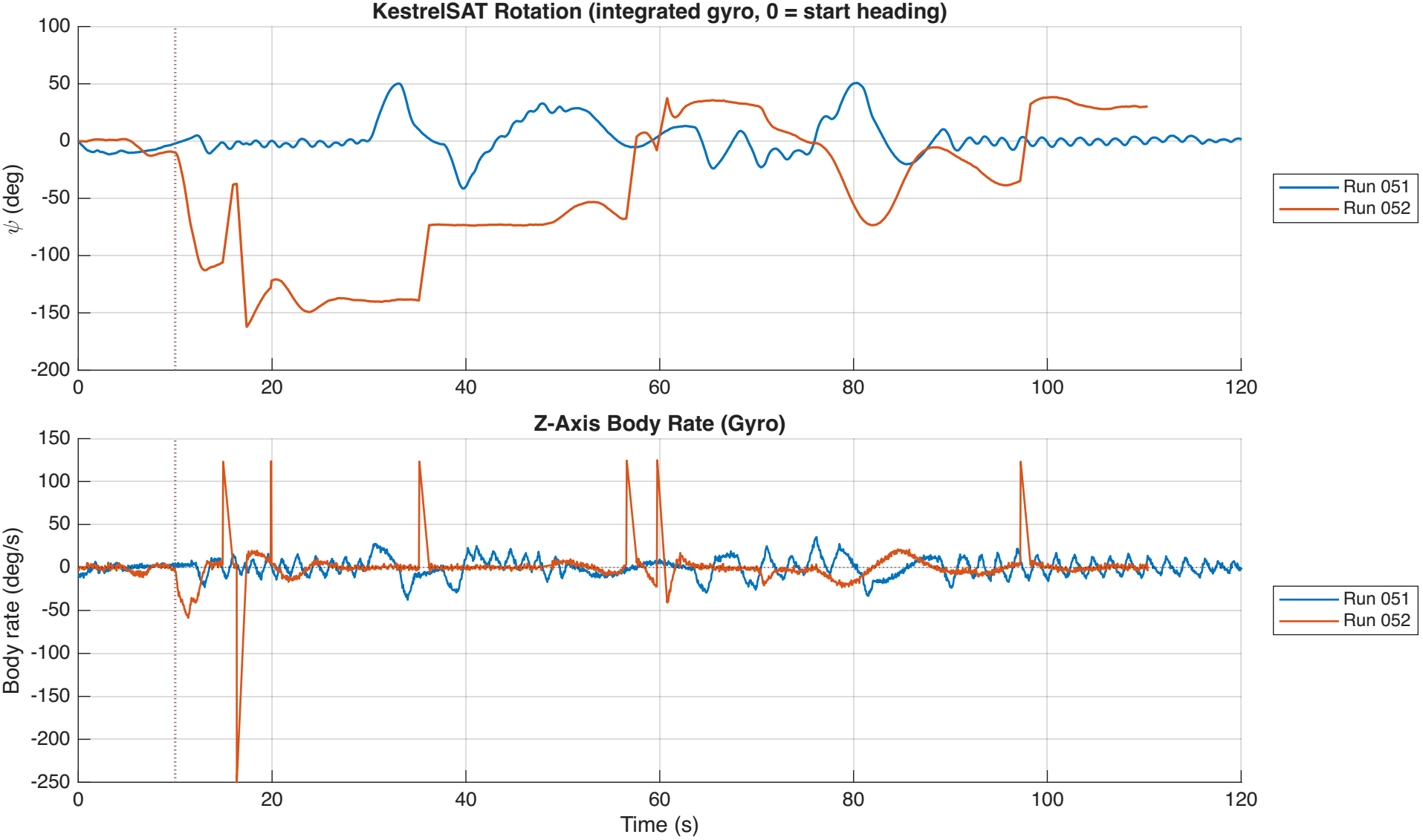
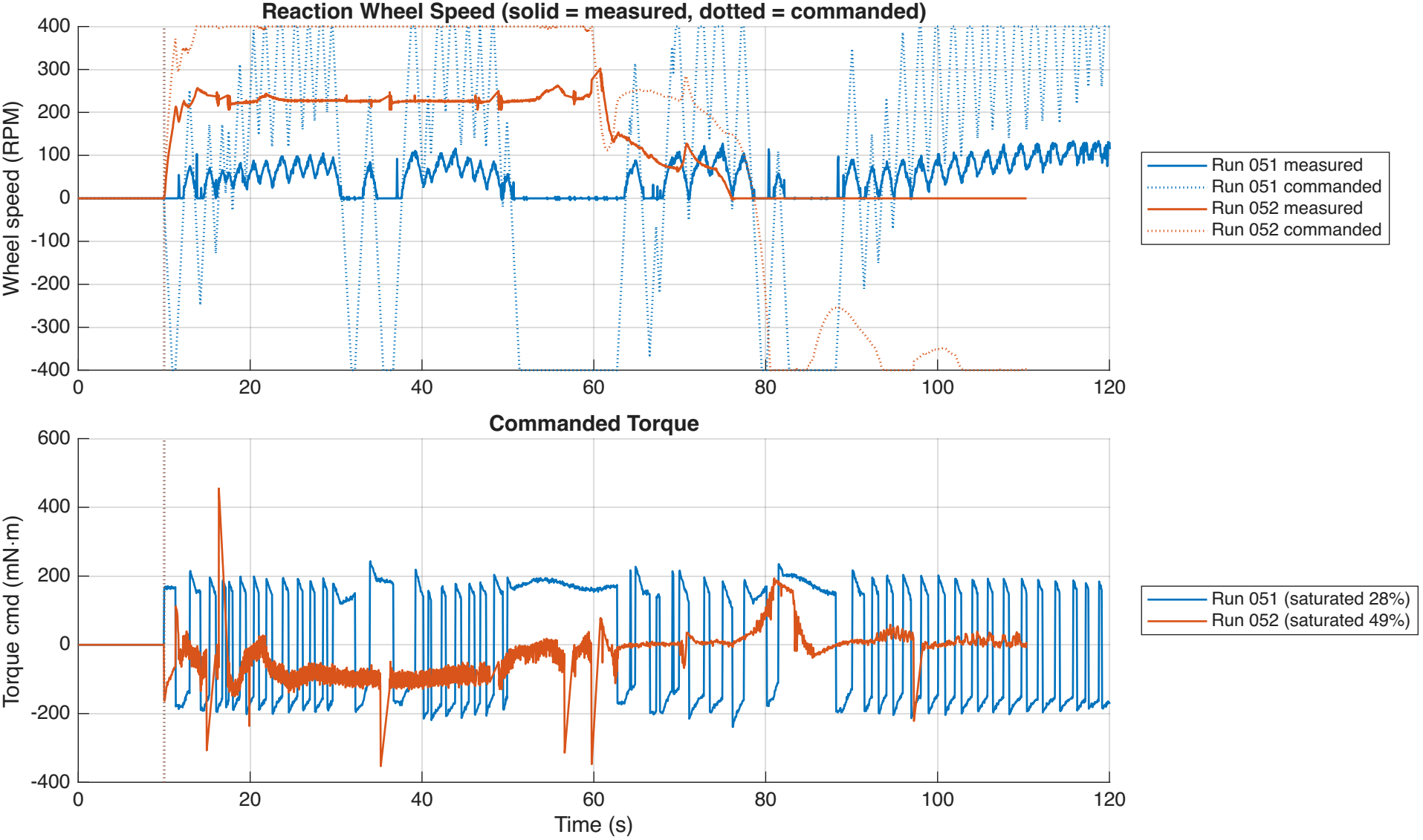
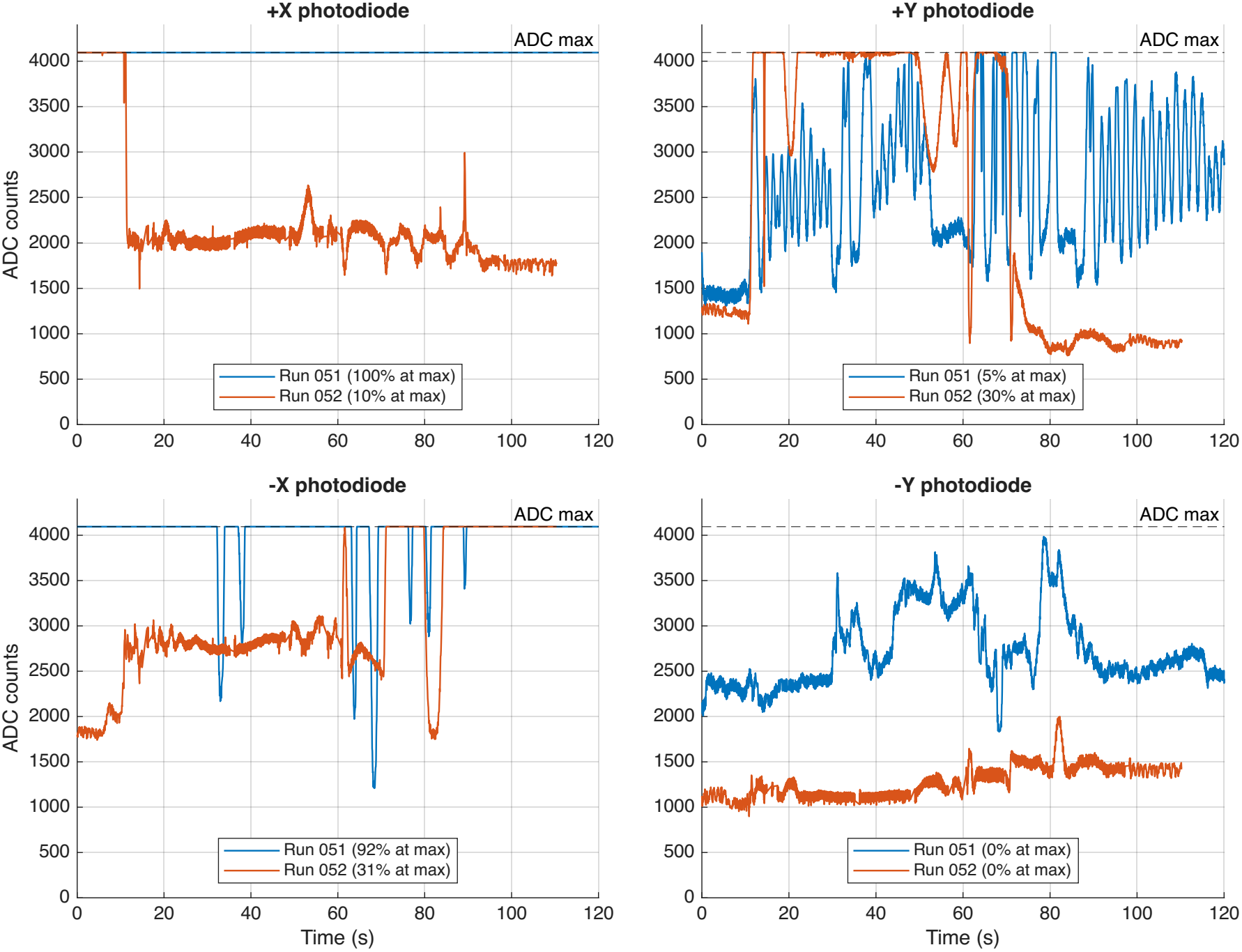
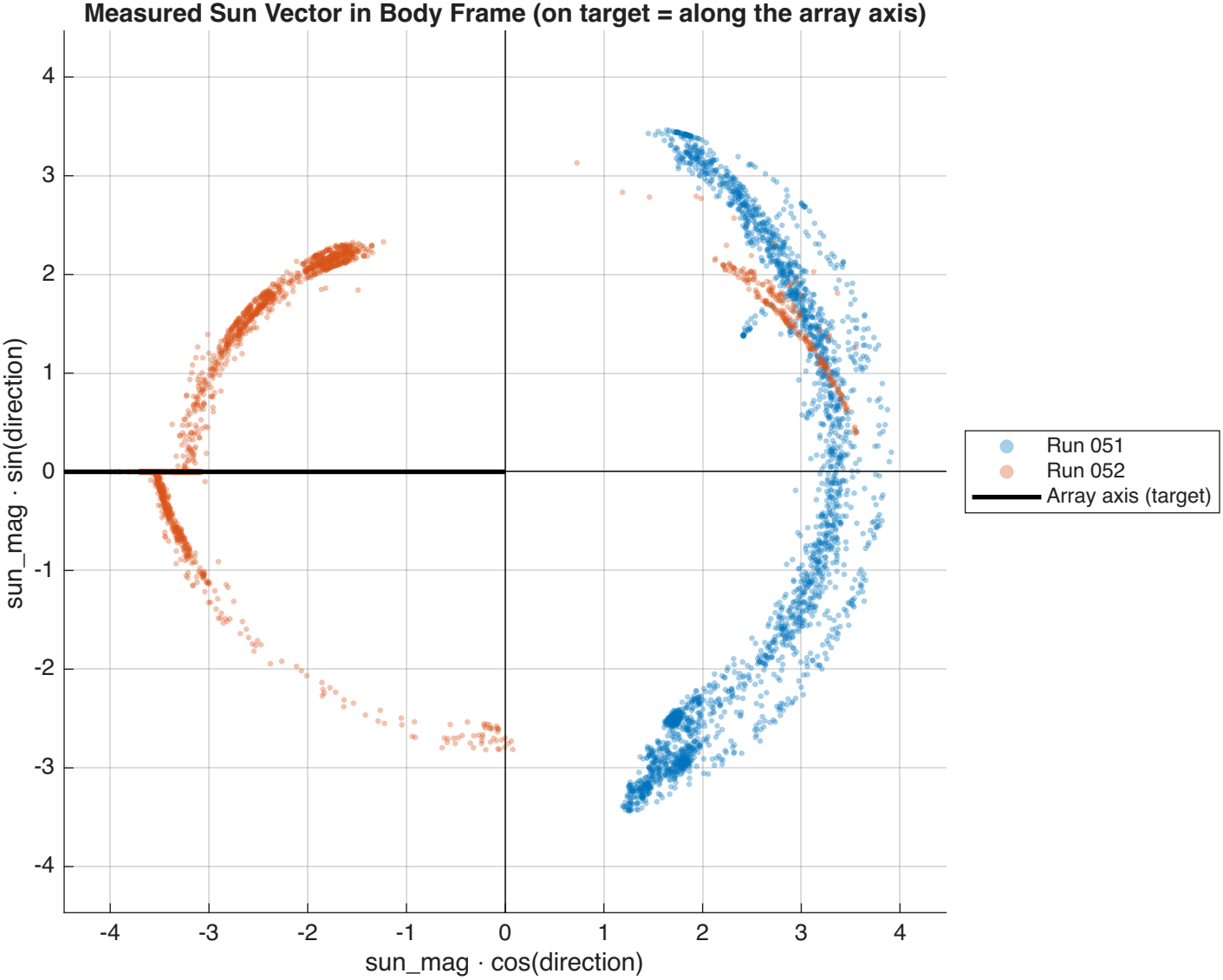
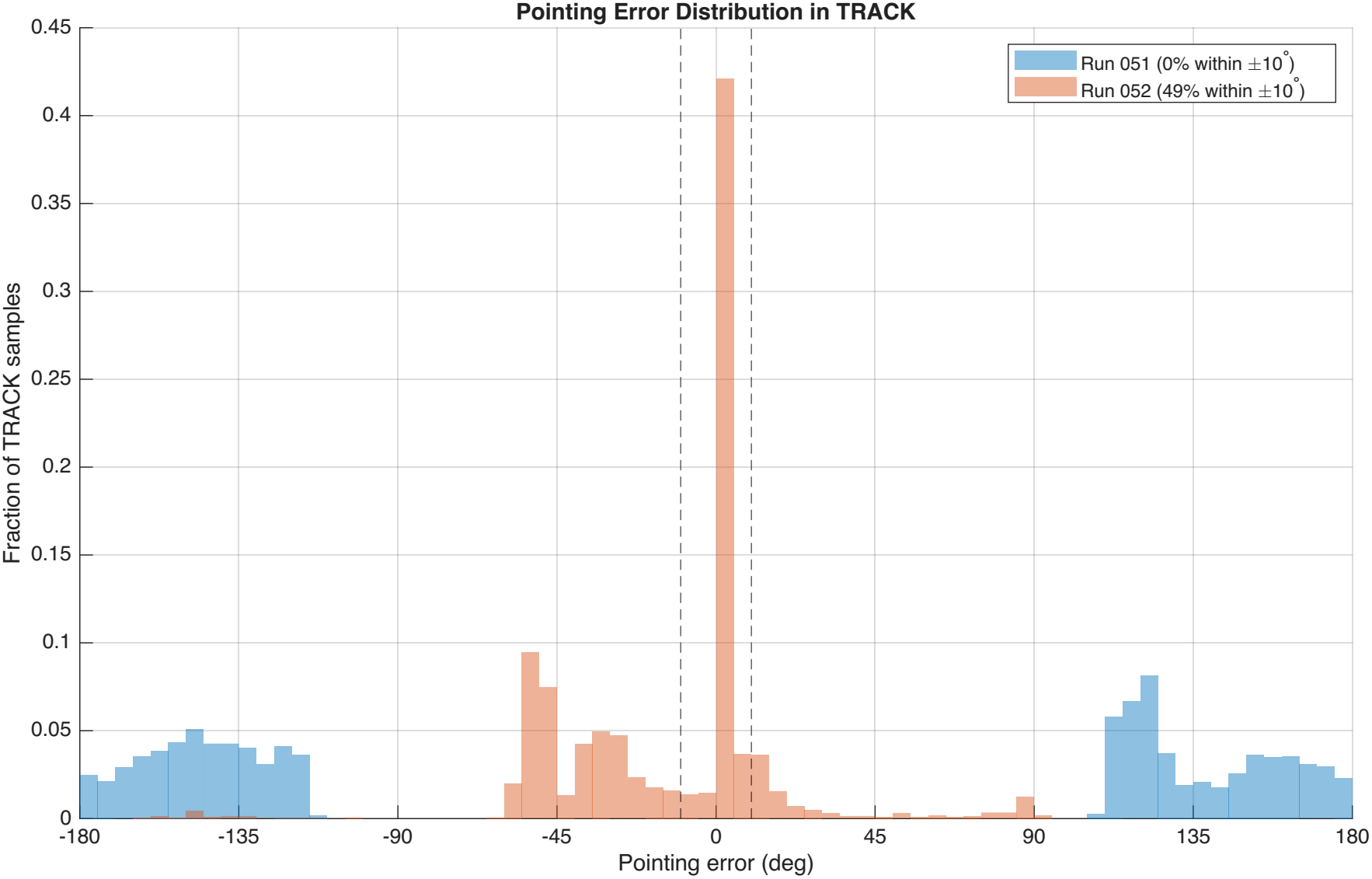

Pointing Error (0 = on target, dashed = ±10°)
Pointing error (deg)
180
135
90
45
0
-45
-90
-135
-180
0
20
40
60
80
100
120
Run 051
Run 052
Sun Direction in Body Frame (target = 180, the -X array face)
Sun direction (deg)
360
270
180
90
0
array axis
Time (s)
Run 051
Run 052


KestrelSAT Rotation (integrated gyro, 0 = start heading)
ψ (deg)
100
50
0
-50
-100
-150
-200
0
20
40
60
80
100
120
Run 051
Run 052
Z-Axis Body Rate (Gyro)
Body rate (deg/s)
150
100
50
0
-50
-100
-150
-200
-250
0
20
40
60
80
100
120
Time (s)
Run 051
Run 052


Reaction Wheel Speed (solid = measured, dotted = commanded)
Wheel speed (RPM)
400
300
200
100
0
-100
-200
-300
-400
0
20
40
60
80
100
120
Run 051 measured
Run 051 commanded
Run 052 measured
Run 052 commanded
Commanded Torque
Torque cmd (mN·m)
600
400
200
0
-200
-400
Time (s)
Run 051 (saturated 28%)
Run 052 (saturated 49%)


+X photodiode
ADC max
ADC counts
Run 051 (100% at max)
Run 052 (10% at max)
+Y photodiode
ADC max
Run 051 (5% at max)
Run 052 (30% at max)
-X photodiode
ADC max
ADC counts
Run 051 (92% at max)
Run 052 (31% at max)
Time (s)
-Y photodiode
ADC max
Run 051 (0% at max)
Run 052 (0% at max)
Time (s)


Measured Sun Vector in Body Frame (on target = along the array axis)
sun_mag · sin(direction)
sun_mag · cos(direction)
4
3
2
1
0
-1
-2
-3
-4
-4
-3
-2
-1
0
1
2
3
4
Run 051
Run 052
Array axis (target)


Pointing Error Distribution in TRACK
Run 051 (0% within ±10°)
Run 052 (49% within ±10°)
Fraction of TRACK samples
0.45
0.4
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
-180
-135
-90
-45
0
45
90
135
180
Pointing error (deg)